TASK SPECIFICATION: P4-01 PHASE P4 — FINANCIAL LEDGER & TUITION OFFSET ENGINE

Financial Ledger & Wallet Database Schema

Git Branch: `feature/P4-01-ledger-schema` · Target Pull Request: feature/P4-01-ledger-schema

Core Objective & Technical Scope:
Append-only double-entry financial ledger schema and student wallet tables with mathematical constraints guaranteeing non-negative balances.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P4-01-ledger-schema`
Upstream Dependencies	P2-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V4__wallet.sql migration creating wallet and ledger_entry tables.
2. Apply CHECK (balance >= 0) constraint on wallet balances.
3. Implement optimistic locking (@Version) on Wallet entity to prevent race conditions.
4. Create WalletRepository and LedgerEntryRepository interfaces.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V4__wallet.sql`
- `wallet/domain/*`
- `wallet/infra/*`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Negative balance operations physically rejected by PostgreSQL engine.
- Idempotency key uniqueness constraint enforced on all financial transactions.

VERIFICATION & TEST COVERAGE STRATEGY

- WalletRepositoryIntegrationTest
- PostgreSQL constraint validation tests